PHISHING EMAIL ANALYSIS REPORT

Cyber Security Internship – Task 2

Prepared by	Aritra Pal
Analysis date	06 November 2026 (sample email date)
Project type	Educational phishing-awareness and email-evidence analysis
Evidence reviewed	Simulated email text, sender notes, URL notes, and one separate .eml file

Important evidence note: The supplied suspicious college email is a fictional training sample. The separate .eml file is a Reddit notification with authentication results that pass; it is not the header source for the fictional college email. These two items are analyzed separately and are not combined into a claim that the Reddit message is phishing.

CONFIDENTIALITY: This report masks the recipient email address and avoids reproducing unnecessary personal identifiers.

1. Executive Summary

This report reviews a simulated college-account security alert and the supporting notes supplied for sender and URL analysis. The training sample uses an urgent suspension warning, an unverified sender identity, and a link requesting account verification. These are common phishing-style indicators. The URL and sender evidence are not sufficient to establish that a real attack occurred.

Overall assessment: The fictional college email should be treated as a suspicious phishing simulation for training purposes. Do not open its link. No evidence supplied demonstrates that an account was actually compromised, that credentials were collected, or that money was transferred.

A key limitation is that the supplied .eml file belongs to a different message: a Reddit notification about a personal-finance discussion. Its SPF, DKIM, and DMARC results pass for the domains shown in that message. Those results cannot authenticate or disprove the separate fictional college email.

2. Objective and Scope

Objectives

• Identify suspicious sender and message characteristics.
• Inspect the URL string without visiting it.
• Review available email-authentication evidence and explain its scope.
• Identify urgency, fear, impersonation, and other social-engineering indicators.
• Record limitations and provide safe handling recommendations.

Scope and evidence set

The analysis is based only on four user-provided files: *phishing_sample.txt*, *sender_analysis.md*, *url_analysis.md*, and a separate Reddit notification in .eml format. No live website was visited, no attachment was executed, and no external reputation lookup was performed. Therefore, this report does not claim to confirm malicious hosting or compromise.

3. Evidence Inventory

Evidence item	What it contains	Use in this report
phishing_sample.txt	Fictional college security alert dated 06 Nov 2026	Primary message-content sample
sender_analysis.md	Sender identity notes and inconsistent SPF/DKIM/DMARC entries	Sender observations; authentication notes treated cautiously
url_analysis.md	Suspicious URL, HTTP protocol, and analyst notes	URL-string inspection
Separate .eml file	Reddit notification; Received and Authentication-Results headers	Separate header-analysis example only

4. Sample Email Details

From: security-alert@college-verify.example
To: Recipient address redacted
Subject: URGENT: Your College Account Will Be Suspended!
Date stated in sample: 06 November 2026
Claimed sender identity: College IT Security Department

Message summary

The message claims unusual login activity on a college account, says the account has been flagged for security verification, asks the recipient to follow a link, and warns that failure to act within 24 hours may cause temporary

account restrictions.

> The sample file explicitly describes itself as fictional and intended for cybersecurity awareness and phishing-detection testing.

5. Sender Analysis

Field	Supplied observation	Assessment
Display name	Security Support	Generic authority-style display name; not proof of identity
Sender address	security-alert@college-verify.example	Unverified; verify through a known college channel
Domain note	Notes list example.com	Does not match the sender domain shown; discrepancy needs clarification
Trust status	Not established / suspicious	Appropriate pending independent verification

The sender-analysis note contains a domain value of *example.com*, while the sample's From address uses *college-verify.example*. This is a mismatch in the supplied notes, not a verified technical finding. The .example suffix is reserved for documentation and the .test hostname used in the URL is reserved for testing; these are consistent with a constructed training scenario.

6. Email Header Analysis

6.1 Header evidence attached to the simulated college email

> No raw RFC 5322 / .eml headers for the fictional college security-alert message were supplied. Consequently, SPF, DKIM, DMARC, Return-Path, Received path, and Message-ID for that sample cannot be independently assessed from the provided evidence.

6.2 Separate .eml file actually supplied

From: Reddit <noreply@redditmail.com>
Subject: Interested in Personal Finance? "Previous employer took 401k from paychecks, now found out there is no 401k account, what do I do?".
Date: Fri, 09 Oct 2026 05:07:56 +0000

The separate message's Authentication-Results lines report SPF=pass, DKIM=pass for redditmail.com and amazonses.com, and DMARC=pass for redditmail.com. Its Return-Path and Received lines show Amazon SES delivery infrastructure. These are consistent with the authentication results recorded by the receiving mail provider for that Reddit notification. They do not prove that every link in the message is safe, and they do not apply to the fictional college sample.

6.3 Conflicting notes

The sender-analysis notes list both DKIM PASS and DKIM FAIL, and repeat SPF PASS and DMARC PASS entries. Because the notes do not identify which message or analyzer result each line refers to, they are internally ambiguous. They must not be presented as confirmed results for the college sample. Obtain and analyze the original raw headers of the exact sample if header authentication is required.

7. URL Analysis

URL observed in the fictional sample:

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http://college-security-verify.example-login.test/account/check?id=928471
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Observed characteristics

Indicator	Evidence / result	Interpretation
Protocol	HTTP	Connection is not encrypted in transit; HTTP alone does not prove phishing
Hostname	college-security-verify.example-login.test	Long, official-sounding hostname; not an independently verified college domain
Path	/account/check	Suggests account verification
Query	id=928471	Contains an identifier-like parameter; its purpose is unknown
Domain mismatch	Not established	The evidence does not provide an official college domain for comparison
Reputation / destination	Not checked	No live visit or reputation lookup was performed

The hostname ends in .test, a reserved testing suffix, and the sample is explicitly fictional. The URL should be treated as a non-production training indicator, not evidence of a live malicious site. The supplied URL notes mention possible session hijacking, account takeover, and personal-data collection; these are possible consequences of credential theft in general, not outcomes observed in this evidence.

8. Social Engineering Analysis

Technique	Evidence in message	Why it matters
Urgency	“within 24 hours”	Pressures recipient to act quickly without checking
Fear / threat	“account will be suspended” and “temporary account restrictions”	Creates concern about losing account access
Authority impersonation	Claims to be “College IT Security Department”	Uses an institutional role to encourage trust
Unusual activity pretext	Claims unusual login activity	Creates a plausible reason to click a verification link
Call to action	Asks recipient to review alert using supplied URL	Moves recipient to an unverified destination

9. Attachments and Payload Review

The supplied fictional email text contains no attachment name or attachment metadata. The separate Reddit .eml file is an HTML email containing ordinary newsletter-style links and unsubscribe metadata in the visible header/body excerpt. No attachment was identified in the provided evidence. No file was executed or opened as an attachment during this analysis.

10. Findings and Risk Assessment

Finding	Evidence strength	Assessment
Urgency and threat language	Directly present	Strong phishing-style indicator
Unverified sender identity	Directly present in supplied notes	Requires independent verification
Verification link in message	Directly present	Suspicious call to action; do not open

Finding	Evidence strength	Assessment
HTTP URL	Directly present	Security concern, but not proof on its own
Authentication of college sample	Not available	Cannot determine from supplied .eml
Actual compromise or data theft	No evidence	Not established

Overall training assessment: Suspicious / phishing-like message. This is a qualitative assessment of the simulated content, not a verdict about a live sender, server, or victim account.

11. Security Recommendations

• Do not click the URL or reply to the suspicious message.
• Verify account alerts by opening the institution's official website independently or contacting IT through a known phone number or service desk.
• Never enter passwords, OTPs, recovery codes, or personal details after following an unsolicited email link.
• Report the message to the institution's security team and preserve the original email as an .eml file for authorized analysis.
• For header analysis, obtain the raw headers from the exact message being investigated and check SPF, DKIM, DMARC, Return-Path, and Received fields together.
• If credentials were entered, change them using the official site, enable MFA, and notify the institution's IT/security team.

12. Conclusion

The fictional college alert demonstrates common phishing techniques: urgency, fear, claimed institutional authority, and a request to verify an account through an untrusted link. These characteristics justify treating the sample as suspicious in an awareness exercise. However, the evidence does not establish that a real website was hosted, that the link was visited, or that any account was compromised. The separate Reddit .eml file shows passing SPF, DKIM, and DMARC results for that separate notification and must not be used as the header result for the college sample.

13. Tools and Methodology

Tools evidenced or applicable to this review: text editor for inspecting supplied text; an email client or raw .eml parser for reading headers; and a trusted email-header analyzer for the raw headers of the exact message. No header-analyzer output for the fictional college sample was included in the evidence bundle. No live URL scanner result is claimed in this report.

14. Evidence Limitations

1. The primary college email is explicitly fictional.
2. The supplied .eml file is a different Reddit message, not the primary sample.
3. No raw headers for the fictional college email were supplied.
4. Sender notes contain conflicting authentication entries and a domain-label discrepancy.
5. URL reputation, DNS records, hosting, redirects, and page contents were not tested.
6. No evidence confirms session hijacking, account takeover, money laundering, or personal-data collection.

15. Interview Preparation – Key Questions

What is phishing? A deceptive attempt to trick people into revealing information or taking unsafe actions through email, messages, or websites.

How can a phishing email be identified? Check sender identity, unexpected requests, urgency, links, attachments, and authentication evidence.

What is email spoofing? Forging or manipulating sender identity information so a message appears to come from another source.

Why are phishing emails dangerous? They may lead to credential theft, fraud, malware infection, or unauthorized access.

How do you verify a sender? Compare the address and domain with a known official contact and review authentication headers for the exact email.

What tools analyze headers? Google Admin Toolbox Messageheader and other reputable header analyzers.

What should you do with a suspicious email? Do not interact with its links or files; report it and preserve it for authorized analysis.

How is social engineering used? Attackers exploit trust, urgency, fear, curiosity, and authority to influence user decisions.

16. Appendix – Evidence Reference

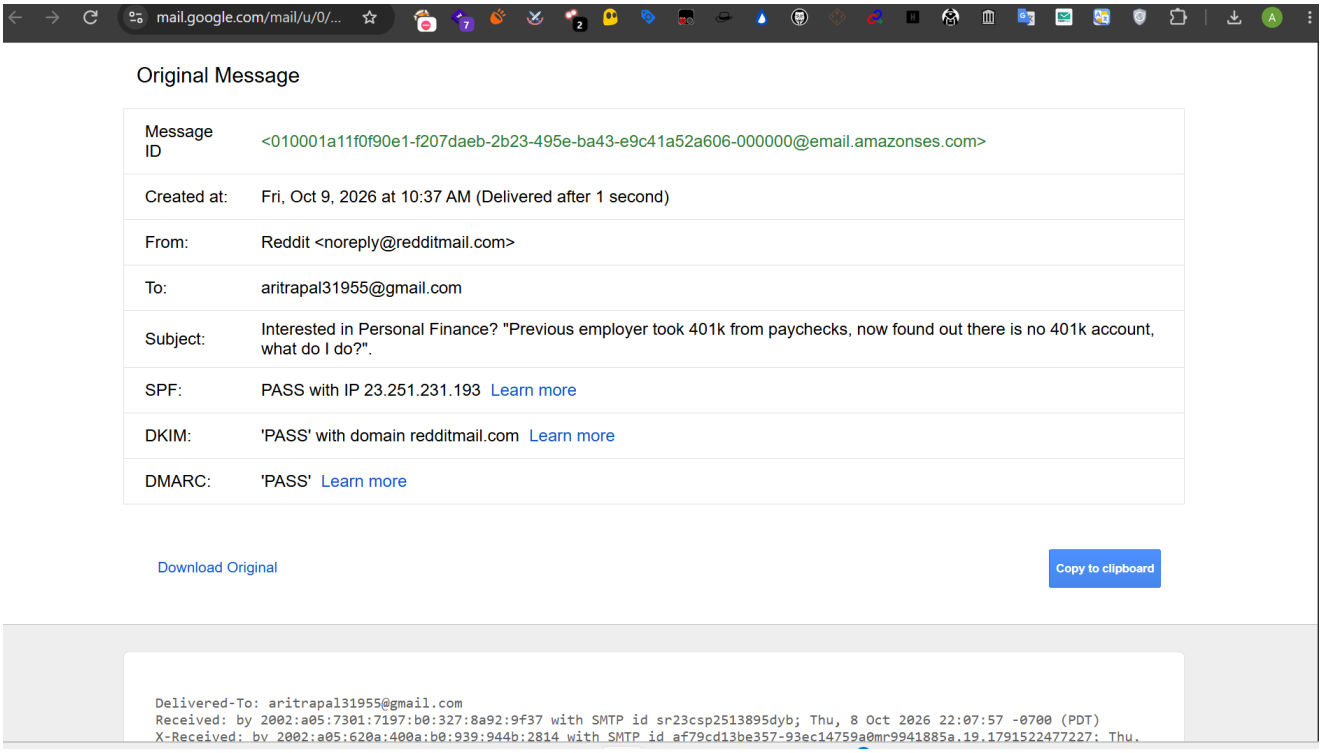
• Primary simulated email: phishing_sample.txt
• Sender observations: sender_analysis.md
• URL observations: url_analysis.md
• Separate email-header sample: supplied Reddit notification .eml file
• Recipient email address has been omitted from this report for privacy.

Disclaimer: This report is prepared for educational cybersecurity awareness and internship documentation. It is based on the supplied evidence and does not constitute a forensic certification or confirmation of a real cyber incident.

Appendix – Supporting Evidence

Screenshot evidence supplied for the phishing-email analysis

Evidence Screenshot 1 – Gmail original-message header and authentication summary

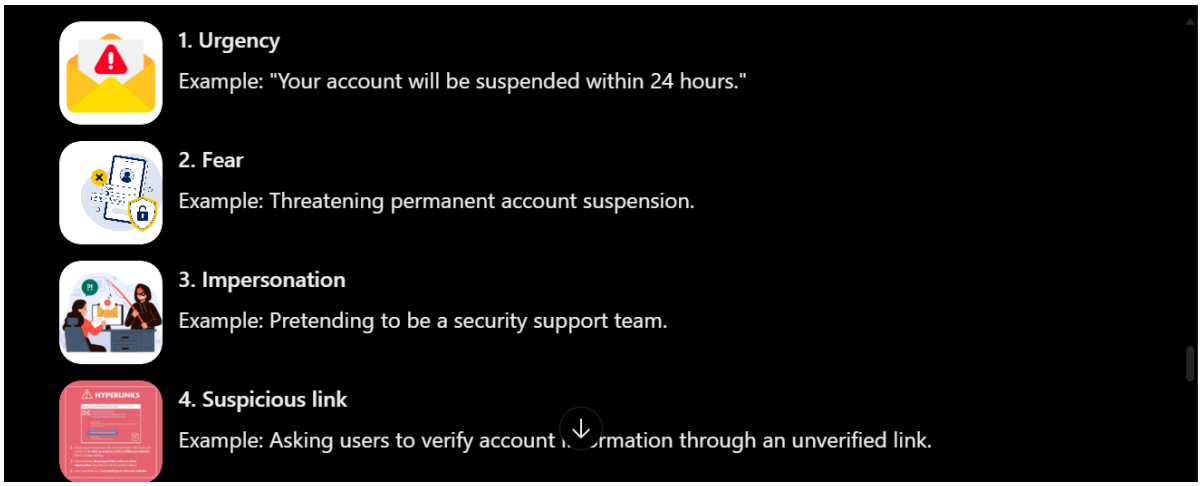


Note: This Gmail header screenshot is for a separate Reddit notification, not the simulated college phishing email.

Appendix – Supporting Evidence (continued)

Visual evidence for social-engineering observations and findings

Evidence Screenshot 2 – Social-engineering indicators



Evidence Screenshot 3 – Findings and risk assessment table

Finding	Evidence	Risk
Urgent language	24-hour deadline	High
Threatening message	Account suspension	High
Unverified sender	Identity not established	Medium
HTTP link	No HTTPS	Medium
Account verification request	Requests account action	High

Screenshots are reproduced as supplied; risk ratings shown are the project's illustrative assessment.